Mind Mirror: AI-Based Webcam System for Stress Detection and Digital Well-Being

A Project Report

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by

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CERTIFICATE

This is to certify that the report entitled **Mind Mirror: AI-Based Webcam System for Stress Detection and Digital Well-Being** submitted by **Advaiy Praveen** (KTU Reg. No: MBT22CS010), **Gaudhami Sasi** (KTU Reg. No: MBT22CS044), **M B Sivanandana** (KTU Reg. No: MBT22CS065), and **Siddharth Rajesh** (KTU Reg. No: MBT22CS107) to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the project work carried out by them under our guidance and supervision. This report has not been submitted to any other University or Institute for any purpose whatsoever.

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Abstract

The increasing reliance on digital devices has led to rising concerns regarding stress, reduced attention, and poor digital well-being among users. Existing solutions often depend on intrusive wearable sensors, cloud-based processing, or self-reported data, which introduce privacy risks, latency, and limited real-world applicability. To address these challenges, this project presents MindMirror, an AI-driven, non-invasive system for real-time stress detection and digital well-being monitoring using a standard webcam. The proposed system integrates Facial Emotion Recognition (FER) and remote photoplethysmography (rPPG) to perform multimodal analysis of emotional and physiological signals. Facial expressions are analysed using FER library and MTCNN, while heart rate is estimated from subtle skin colour variations in video frames. These signals are fused to infer stress levels accurately and continuously. The system operates entirely on-device, ensuring user privacy and eliminating the need for external hardware or cloud infrastructure. Based on real-time analysis, it provides intelligent interventions such as stress alerts, to promote healthier screen usage. Experimental implementation demonstrates the feasibility of a scalable, cost-effective, and privacy-preserving solution for continuous stress monitoring in real-world environments. By combining AI-driven analysis with actionable feedback, MindMirror contributes to enhancing mental well-being, improving focus in digital environments, and supporting sustainable human-computer interaction.

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Chapter 1

Introduction

Nowadays, technology is taking over our lives and children, as well as adults, are constantly staring at the computer screens to work, learn and have fun. Actually, they don't get the notion of how it affects their focus, stress levels and digital health in genera. Besides, continuing staring at the screens might cause mental fatigue, inability to pay full attention and stress. Unfortunately, we only notice, face and tackle these problems when they are so severe that they affect the quality of our lives. Most of the methods for measuring user engagement that we have today- for example wearable sensors, self-report questionnaires, or special hardware are very invasive, inconvenient or highly inaccurate due to the problems of biased user responses, and frequent misusage.

MindMirror comes to fill this very important void by offering a method that is totally non-invasive, very easy and natural to the user and it just requires a regular computer camera - no other equipment is necessary. Special deep learning methods used for recognizing facial expressions plus remote photoplethysmography (rPPG), which the system utilizes come together to provide both emotional and physiological information at the same time. Facial expression and heart rate are the main indicators the system uses to keep track of ones mood continuously. It does a great job in detecting that someone might be starting to feel tired from too much screen time, having too much on their plate mentally or getting stressed emotionally which can then be controlled by the timely feedback given by the system. Besides, it is a great tool for promoting healthy digital habits.

Chapter 2

Literature Review

2.1 Video-Based Stress Detection through Deep Learning

The authors built deep learning models to predict stress based on webcam video of 122 college students. They proposed a novel model called TSDNet, a two-stream ResNet + LSTM model that integrated facial features and optical-flow features. Their findings showed that TSDNet classified stress with 85.42% accuracy and 85.28% F1-score and was an improvement over facial-only features and optical-flow only models. They acknowledge the limitation of only using single episodes, or short videos in the lab, and they were not observing individualized stress response. MindMirror is using this work as a foundational study and will be employing a multilayered approach compared to TSDNet in real time for the prediction of stress and provide features to enable behavior mapping in real time to provide personalized adaptive feedback to the user. [1]

2.2 Comprehensive Review and Analysis on Facial Emotion Recognition

They provide a systematic comparison of interpreting traditional machine learning classifiers against deep learning models and their hybrids (i.e. convolutional neural network [CNN] with support vector machine [SVM] pipelines) achieving 99% accuracies in FER datasets. They also illustrate the importance of data preprocessing, data augmentation, and feature fusion from deep learning and traditional models in achieving high-performance FER targets. MindMirror builds upon these observations highlighting hybrid and two-stream approaches achieve high-performance emotion classification; the hybrid and two stream approaches also provide architectural flexibility to the algorithms used in the real world and varied lighting conditions, and varying poses. [2]

2.3 RGB Camera-Based Physiological Sensing: Challenges and Future Directions

The investigators observed advancements in non-contact rPPG sensing. Although they see advancement in signal recovery from using advanced CTNs (deep learning CNN), the researchers noted that either way, the bias was not thoroughly addressed or corrected accounting for skin color bias and non-invariance changes to lighting conditions remain. MindMirror features improvements via adaptive normalization associated with rPPG, multimodal–stress fusions, and equity processes for a better reliable assessment for stress in user population. [3]

2.4 Student Engagement Assessment Using Multimodal Deep Learning

The authors of the evaluation of the evaluation of student engagement employed their ResNet-50 CNN which created the model for measuring engagement based webcam images of the students. The CNN was able to classify engagement with 91 percent accuracy for stress detection, but it is important to note that these classifications were solely based on visual gesture cues. The main study did not include physiological validation or real-timed interactivity. MindMirror builds on this work by assessing heart-rate signals derived from rPPG and facial feature data to create a multimodal interface for estimating stress, and also includes real-timed interactivity around the feedback systems. [4]

2.5 Detection of In-the-Wild and Long-Term Stress Based on Cross-Attention Transformer and Context-Aware Ensemble Model

The researchers created a model for stress detection for maritime controllers using Cross-Attention Transformers that delivered 92.12% adaptive accuracy with remarkable context adaptability, but used a custom hardware sensor. They wanted to eliminate the hardware requirement. MindMirror tackles this roadblock, and thus achieves a similar standard of context adaptability with a similar number of personalized data and behavioral feedback loops simply utilizing a webcam. [5]

2.6 Lightweight Advanced Deep-Learning Models for Stress Detection on Social Media

The authors developed a CNN-based FER model supported by a Flask app for live observation. Although it achieved 98 percent accuracy overall, it had no integration of physiological signals or a variety of training data. MindMirror builds on these ideas by combining FER and rPPG data streams and providing personalized recommendations within a privacy-preserving architecture. [6]

2.7 Real-Time Stress Detection and Analysis Using Facial Emotion Recognition

Developed a CNN based model trained on the FER-2013 dataset to detect emotions from a webcam feed with 98% test accuracy and trend visualization for stress but did not validate physiology. MindMirror improves upon this approach with rPPG, dynamic stress tracking, and cognitive well being prompting to create an enhanced user experience. [7]

2.8 Real-Time Mental Stress Detection Using Multimodality

This study used CNN-GRU networks to produce both facial and physiological data. In total, the facial based model had great success in producing meaningful results, however a key limitation of this work is that the model relied on physical sensors which restricts the scalability of the model. MindMirror builds on this study because it allows both modalities to be taken for and extracted solely through webcam. The computer vision based approach is non-intrusive and scalable while also maintaining privacy when engaging in stress monitoring. [8]

2.9 Gaps in the Literature Review/Limitations of Existing Approaches

2.9.1 Video-Based Stress Detection through Deep Learning

Although the TSDNet model performed well at achieving satisfactory accuracy, this study was limited to short lab-recorded space videos in a controlled environment. Therefore, the generalizability of our model and results may be decreased in uncontrolled real-world environments where light, motion, or other environmental conditions change. In addition, the model was a generalized program that model could not adapt to individuals' unique stress expression or behavioral habits in stress detection, limiting its potential effectiveness for continuous real-time monitoring of stress.

2.9.2 Comprehensive Review and Analysis on Facial Emotion Recognition

Although this paper made an effective comparison of traditional vs. deep learning-based emotion recognition models with very high accuracy, the scope was limited to benchmark FER datasets, which are generated and gathered under limited conditions, and not for spontaneity or situational relevance. Moreover, physiological signals, as well as personalization factors, were not offered or accounted for in assessing accurate stress detection beyond visible emotion cues.

2.9.3 Camera-Based Physiological Sensing: Challenges and Future Directions

This review highlighted the progress made in non-contact physiological sensing, using rPPG methods, yet also identified critical issues such as, bias under different skin types, light conditions, and camera quality. The lack of standardization of datasets and their diversity and variability, and the particularity of deep learning model classification, being sensitive to environmental factors in deployment tools, make these advances less applicable in real-world, unconstrained conditions. It is also noted that personalization and adaptive calibration across users was not examined extensively.

2.9.4 Student Engagement Assessment Using Multimodal Deep Learning

Although the research demonstrated the ability to measure engagement accurately, individual differences in stress levels due to context, and change in mood were evident however it did not use body signals that would make it possible to detect even the slightest and most objective psychological stress. In addition, a disadvantage of the study was the lack of real-time feedback and interactivity, thus limiting practical applications within classrooms or mental health settings. Furthermore, the lack of multimodal fusion limits the model's robustness and interpretability for individuals with varied backgrounds.

2.9.5 Detection of In-the-Wild and Long-Term Stress Based on Cross-Attention Transformer and Context-Aware Ensemble Model

The current study exhibited a powerful deep learning model that could learn adaptability to the stress based-contextual factors. However, the learning in these experiences instead classified on specialized hardware sensors for physiological stimuli and data collection, and therefore less scalable (less practical) for acquisition outside controlled, industrial environments. Dedicating the use of specified sensors for this models only increases a user's discomfort and accessibility. In addition, the cost of stress acquisition and monitoring would also limit a universal use for everyday purposes and stress monitoring.

2.9.6 Lightweight Advanced Deep-Learning Models for Stress Detection on Social Media

While the authors achieved extremely high accuracy with a CNN-based FER model deployed as a real-time web application, they relied purely on emotional cues from images harvested from social media. The lack of physiological or multimodal data use potentially led to superficial inferences about stress and marginally addressed contextual accuracy. Further, the lack of diversity in the FER-2013 dataset created potential biases and could have limited generalizability across demographic groups.

2.9.7 Real-Time Stress Detection and Analysis Using Facial Emotion Recognition

Despite achieving good test accuracy with the FER-2013 dataset, the study's stress detection model used visual emotion recognition to detect stress and did not include the validation of the model through physiological metrics (e.g., heart rate or rPPG). The lack of physiological data again limited the model's ability to detect subtle and chronic signals of stress. Even though the study offered visualizations of their organizations' stress trends, they did not incorporate adaptive learning or feedback mechanisms to offer an overall longer-term personalized stress management system.

2.9.8 Real-Time Mental Stress Detection Using Multimodality Expressions with a Deep Learning Framework

This research demonstrates an effective fusion of facial and physiological modalities through CNN-GRU architectures; however, it relied on physical sensors, such as ECG or GSR devices, which restrict scalability and user convenience. The use of wearable or contact-based sensors limits monitoring in real-time and continuous monitoring in everyday contexts. Moreover, the framework did not incorporate privacy or personalization, which are important for user trust and widespread usage.

Chapter 3

Proposed Methodology

3.1 Problem statement

The lack of efficient, non-invasive, and intelligent real-time systems for monitoring mental well-being leads to undetected stress and poor digital habits, highlighting the need for an AI-driven, webcam-based multimodal solution that provides accurate and continuous stress detection using facial and physiological signals.

3.2 Proposed Methodology

The proposed system, MindMirror, is a fully non-intrusive artificial intelligence framework capable of evaluating states of stress, engagement, and digital fatigue through the recognition of facial expressions (FER), and remote photoplethysmography (rPPG) monitored through a standard webcam while using a computer screen. The system monitors facial micro-expressions and subtle physiological changes indicating heart rate to continually and discretely assess a user's cognitive and emotional states using a multimodal fusion engine. A novel feature of MindMirror is that it representation both an emotional state, physiological state, and behavioral state form an holistic sensor-free assessment of digital well-being. The system tailors itself to each user through personalized baseline thresholds and prompts a break or relaxation suggestion after consenting threshold levels of sustained stress have been observed. Over prolonged period of time of usage in, MindMirror builds a digital personality map that illustrates the effects of different online spaces and related user's behavior on attention, mood, and stress patterns.

The proposed system consists of the following modules:

3.2.1 Data Acquisition Module

This module captures continuous video streams in real-time from the webcam using OpenCV. Frames are captured at a rate of 15-30 FPS and then processed for recognizing facial expressions (FER) and extracting remote photoplethysmography (rPPG). The patterns of user interaction, such as which background applications are active or user's screen activity, are logged and analyzed for possible behavioral associations.

Finally, the module ensures that only stable, well-light and usable frames are produced for subsequent detailed analysis.

3.2.2 Data Preprocessing and Face Detection Module

The captured frames are essential for establishing the base of standardization. At this step, the original frames will be resized, adjusted to a standard brightness level, and transformed from RGB to grayscale. To detect faces, the process utilizes the MTCNN as the principle face detection method, since the method has fidelity in detecting key facial landmarks. In the case of multiple faces, the module chooses the region of interest (ROI) face with the highest confidence score and bounding box. The selected face is cropped, aligned, and optimized for FER and rPPG processes; only delivering clean, quality facial data to the next module.

3.2.3 Facial Emotion Recognition (FER) Module

This Module will apply a convolutional neural network (CNN) based deep learning model, for example a TSDNet or FER-CNN model, in predicting emotions in response to happy, sad, angry, fears, surprise, and neutrality expressions. Inputs from previous module for detecting facial landmarks should improve stability and accuracy in detecting facial expressions. Processed frame outputs will be associated with probability scores for ALL prediction emotions, aggregating a moment to moment timeline of emotional responses for each instance. This continuous insight in emotional predictions should contribute toward measuring user engagement and inferring stress level.

3.2.4 Heart Rate Estimation (rPPG) Module

Either the CHROM or PyVHR algorithm may be utilized in this module to gauge heart rate based on slight color variations in the user's facial skin regions. A bandpass filter (0.7–3.0 Hz) eliminates pulse-related frequency components into the ambient noise. The resulting BPM value is normalized using a moving-average filter to reduce fluctuations. This method's lack of contact means the user may remain comfortable without any wearable sensor.

3.2.5 Fusion and Stress Detection Engine

The fusion method first combines the emotion probabilities with the trends of the heart rate into a single composite measure of the stress. In a rules-based system for example a negative emotion or heart rate beyond a preset limit every frame is considered stress or not stress. This multimodal fusion indeed makes the system more robust since the validation is done through multiple channels of signals and we may even switch to adaptive fusion methods based on machine learning for more accurate performance.

3.2.6 Real-Time Emotion and Engagement Monitoring

This module logs a person's facial expression recognition (FER) results and physiological modifications to gauge the extent of their engagement. It detects abrupt shifts in emotion, prolonged expression of neutrality, heart rate increases, and rapid changes in attention signals. The analysis of FER outcomes and physiological alterations in real time is employed to detect digital fatigue, distractions, and stress accumulation. Such pinpointing supplies rich context concerning user interaction with digital material.

3.2.7 Live Visualization and UI Dashboard

Created using the Flet framework, the dashboard shows the live webcam video along with the emotion detection, heart rate, and stress state updated in real time. If the duration of stress goes beyond the defined limit (e.g. for 20 seconds) of time, the dashboard automatically produces stress alerts, which help users, most of all, children, to become aware and control their online behaviors.

3.3 Operating Environment

3.3.1 Software Requirements:

Programming Language: Python 3.10 or above.

3.3.1.1 Frameworks and Libraries:

- Flet for GUI and user interface
- OpenCV for video processing and facial detection
- OpenCV for video processing and facial detection
- NumPy, SciPy for signal and data processing
- PyVHR (optional) for advanced rPPG analysis

Development Environment: Visual Studio Code / Jupyter Notebook
Operating System: Windows 10 / 11 (64-bit)

3.3.2 Hardware Requirements:

Processor: Intel Core i5 or higher
RAM: Minimum 8 GB
Storage: Minimum 10 GB free disk space
Webcam: HD (720p or higher)
Optional: GPU (NVIDIA CUDA-compatible) for faster emotion recognition

Chapter 4

Project Design

4.1 High level architecture diagram

MindMirror's architectural layout resembles a modular, real-time AI-driven system, which couples facial emotion recognition (FER) with remote photoplethysmography (rPPG) for monitoring of stress as well as digital well-being via a standard webcam. Initially, the system comes with real-time data acquisition wherein OpenCV is used to capture the live video frames. These frames are normalized, and for face detection, MTCNN is used. After that, the face region is cropped from the frame. The facial pixels are fed into two separate pipelines simultaneously: CNN-based FER which infers the emotion from facial expression and rPPG which calculates the heart rate from subtle color variations of the face. The fusion engine, which applies rule-based or adaptive logic to determine stress levels, combines the outputs of these two. The output of the system is displayed in real-time on a Flet-powered dashboard, where the user can monitor their current mood, heart rate, and stress level. Besides that, the system stores these parameters along with the time feature to allow for the user's long-term analysis, thus creating periodic reports and behavioral insights that constitute the user's Digital Personality Map. The architecture is modular, scalable, and equipped with real-time feedback features, thus, it is capable of accurate, non-intrusive emotional and physiological monitoring of the user.

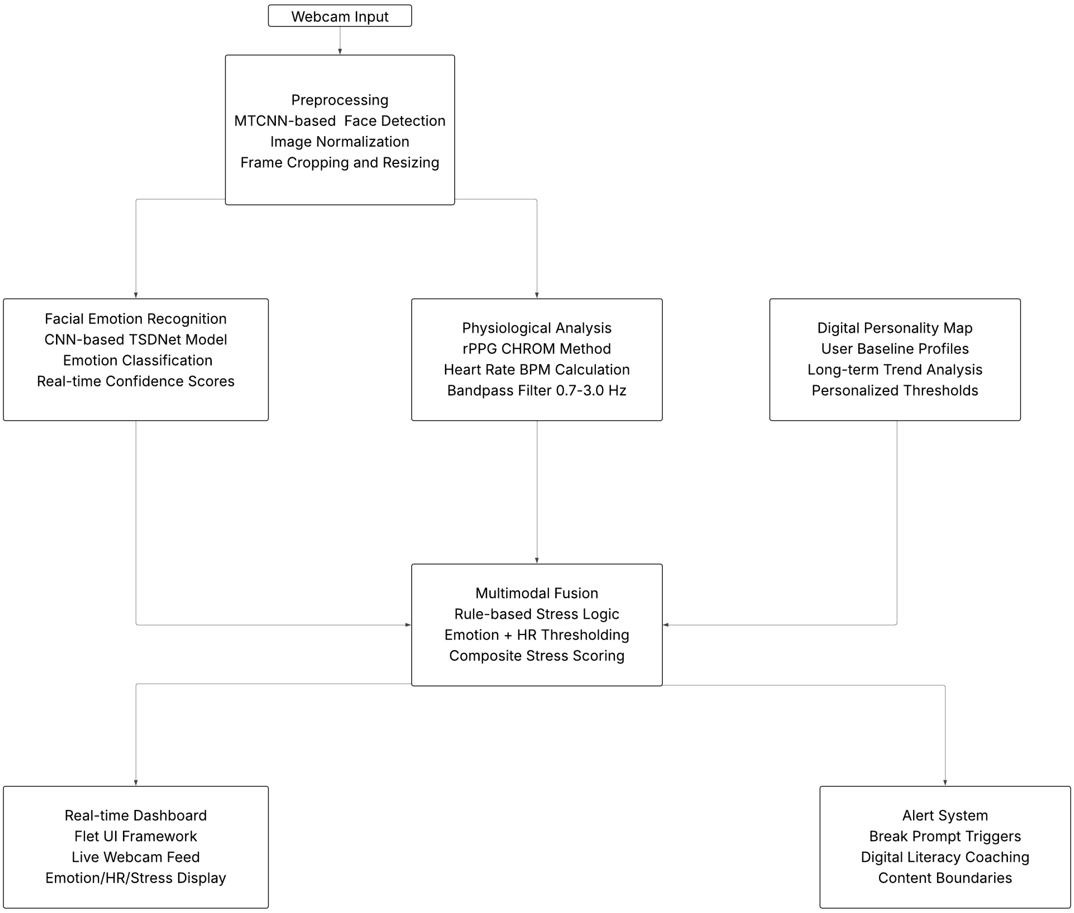

Figure 4.1: Architecture Diagram

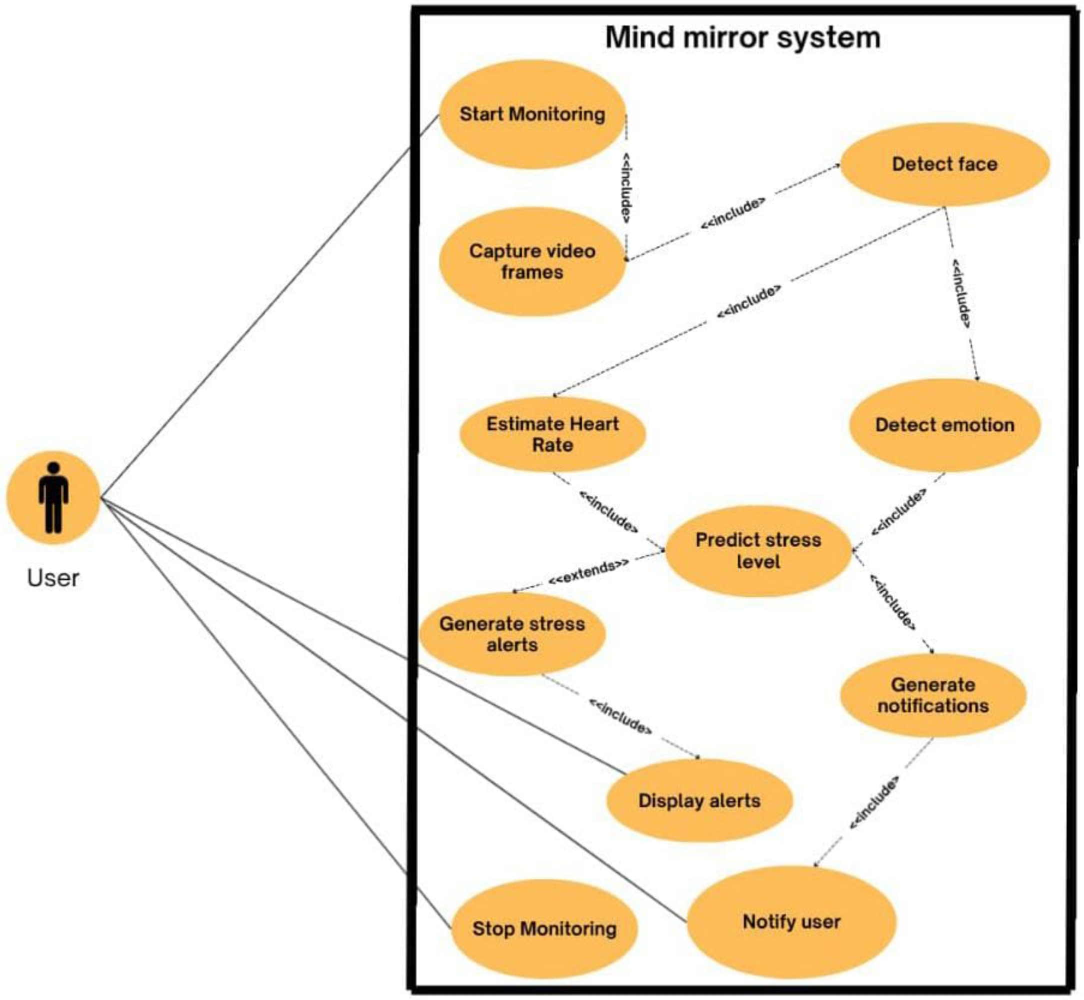

Figure 4.2: Use Case Diagram

4.2 UML diagrams

4.2.1 Use-Case Diagram

The Use Case Diagram represents how the user and the MindMirror system interact with each other and what main functions the system can offer. The user is the primary actor in the system, who starts and manages the monitoring process. The system gets working when the user picks the Start Monitoring option, which initiates the grabbing of video frames with the webcam. These frames go through several interconnected use cases inside the system, including face detection, emotion recognition, and heart rate estimation. Then, by integrating emotional and physiological data, the system makes a stress prediction. Consequently, the system produces the alerts and notifications for the user that are triggered by the stress level forecast.At the same time, there is a mention of the logical relationships such as include and extend in the diagram, which show the dependencies between processes like emotion detection and stress prediction. Eventually, the user ends the session with the system by choosing the Stop Monitoring option, thus closing the interaction loop.

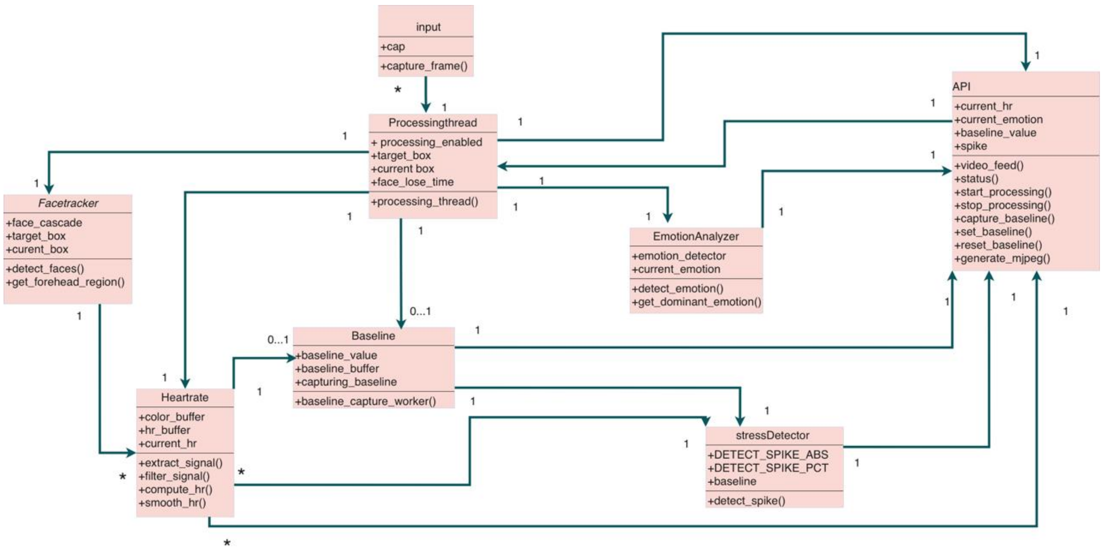

Figure 4.3: Class Diagram

4.2.2 Class Diagram

The Class Diagram gives us the structural design details of the MindMirror system by specifying different classes along with their attributes, methods and relations. The system consists of several interconnected classes including FaceTracker ProcessingThread EmotionAnalyzer, HeartbeatSensor StressDetector Baseline, and API. Each class implements a particular feature. For example, FaceTracker is in charge of face detection and region extraction whereas EmotionAnalyzer interprets facial data for identifying the emotional states of the person. HeartbeatSensor obtains heart rates based on rPPG signals, and StressDetector combines the inputs from the modules to calculate stress levels. Baseline class contains reference values per user to personalize the analysis. ProcessingThread is like the main brain that organizes the data flow between these elements. Class relationships by means of associations and dependencies illustrate how information moves within the system. API class acts as a communication means between the backend processing units and the frontend user interface, guaranteeing the interaction and data exchange seamlessly.

4.2.3 Sequence Diagram

The Sequence Diagram illustrates how system components interact dynamically through time during a typical run of the MindMirror system. The figure shows that the user starts the process by opening the app via a browser interface which sends requests to the backend server (Flask API). After launching, baseline data are acquired and preserved via the baseline manager. Subsequently, the system starts the processing thread that keeps on obtaining camera video frames. These images are simultaneously forwarded to the facial emotion recognition (FER) and heart rate estimation modules. The FER module identifies the emotions whereas the rPPG module measures the heart rate. Then, the outputs go to the stress detection module which determines if the user is under stress. Moreover, the system can update the baseline from time to time and

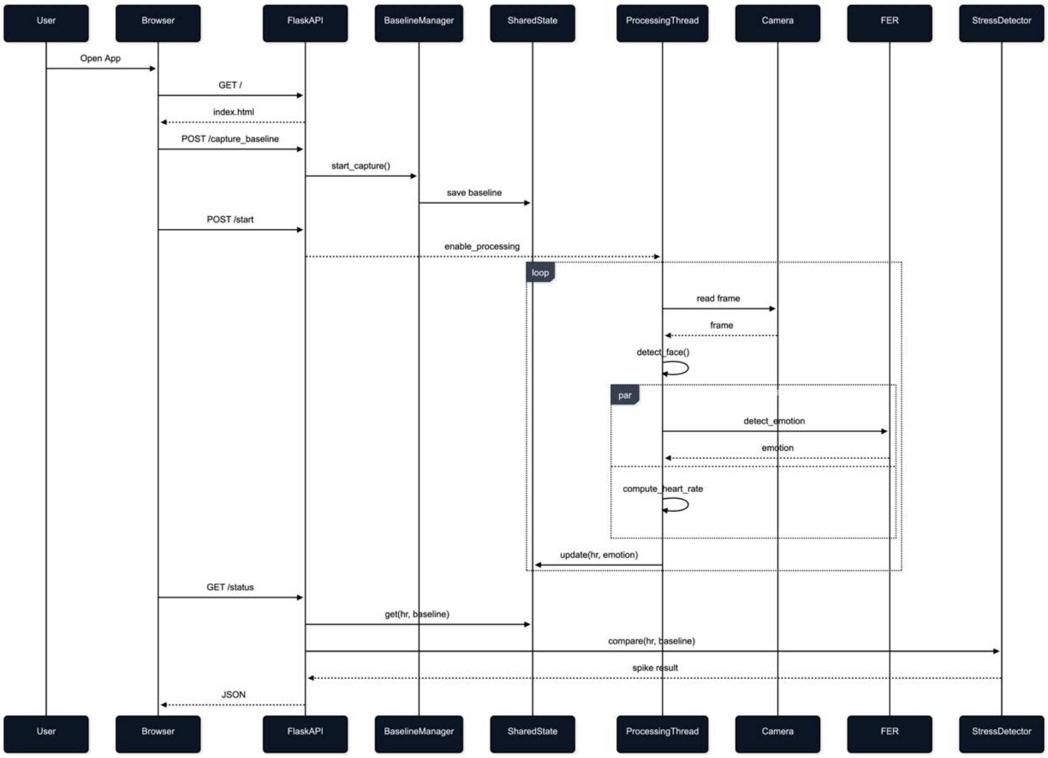

Figure 4.4: Sequence Diagram

calculate the stress level at any moment. Finally, the data after processing are sent back to the frontend formatted in JSON and shown to the user, thus finishing the interaction loop.

4.2.4 Data Flow Diagrams

4.2.4.1 Level 0 diagram

The Level 0 Data Flow Diagram gives a broad overview of the MindMirror system portraying it as one single process. Here the user is the external entity who supplies the system with raw video data captured through a webcam. MindMirror system (showing as Process 0.0) takes this input and sends out stress alerts, detected emotions, and heart rate deviations as outputs. All internal processes have been left out at this level and only the input-output relation is shown, which is very helpful in understanding the logic of the system without getting the internal details.

4.2.4.2 Level 1 diagram

The Level 1 DFD breaks down a system into four main processes: Data Acquisition and Preprocessing (1.0), Multimodal Analysis (2.0), Stress Detection (3.0), and Alert and Warning Output (4.0). First, video frames of the user are taken and preprocessed to get the facial data needed. The processed data is then sent to the multimodal analysis module where the emotional features and physiological signals (heart rate patterns) are extracted. The two outputs are merged and sent to the stress detection module

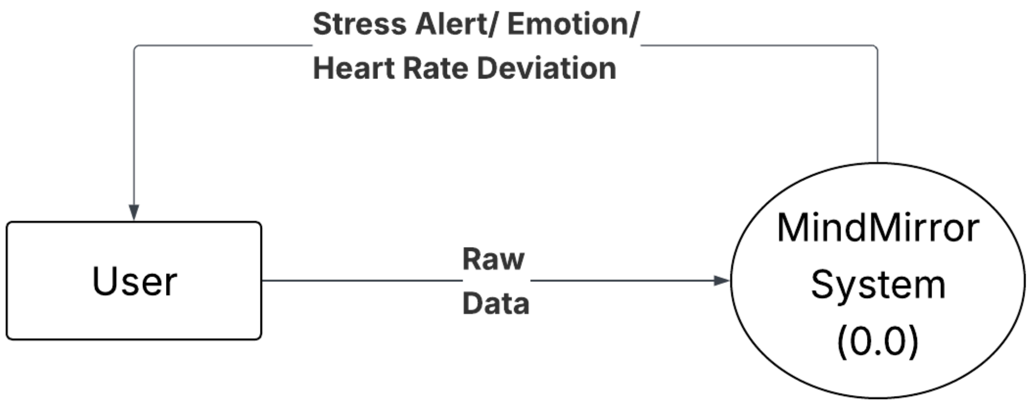

Figure 4.5: DFD Level 0

which determines a stress factor. Based on the final stress level, the alert and warning module creates suitable notifications which are then sent to the user. This chart very well illustrates the sequential data flow through the main functional components of the system.

4.2.4.3 Level 2 Diagrams

Process 1.0 (Data Acquisition and Preprocessing)

The second level of the Data Flow Diagram (DFD), shown in the figure, specifies the internal process of the "Data Acquisition and preprocessing" module (Process 1.0) in detail. The process begins when a user provides video frames as an input to the system. The first step requires the system to obtain the raw frame and then goes through a face detection step to find and extract the area of interest (Face ROI Frames). This step is instrumental in obtaining face data necessary for any subsequent analysis.

Once the Face ROI frames have been extracted, the DFD provides an indication that the preprocessing data will need to go through a "normalization and preparation" step. This is to ensure that the facial data is prepared properly, in a manner in which facial data will be consistent and the data are in a usable format while avoiding issues related to lighting or distance variation. Therefore, the outcome of all of this processing will result in preprocessed data, in whichever data form is normalized Face ROIs, which will then be passed along to the multimodal analysis as the next system module.

Process 2.0 (Multimodal Analysis)

This diagram illustrates the internal workflow of the multimodal analysis module. The process begins with the module receiving preprocessed data from the prior stage. Inside the multimodal analysis module two key analyses are conducted in parallel. First, facial expressions will be analyzed to produce an emotion score, while rppg technology will be used to obtain a heart rate estimate and a corresponding hrv trend. After both analyses are completed, these two separate flows of emotional data and physiological data are inputted into the fusion process. The fusion process collects all features from both streams and generates a unified score. Combined, the output from the multimodal analysis module is a combined set of three metrics that is provided to

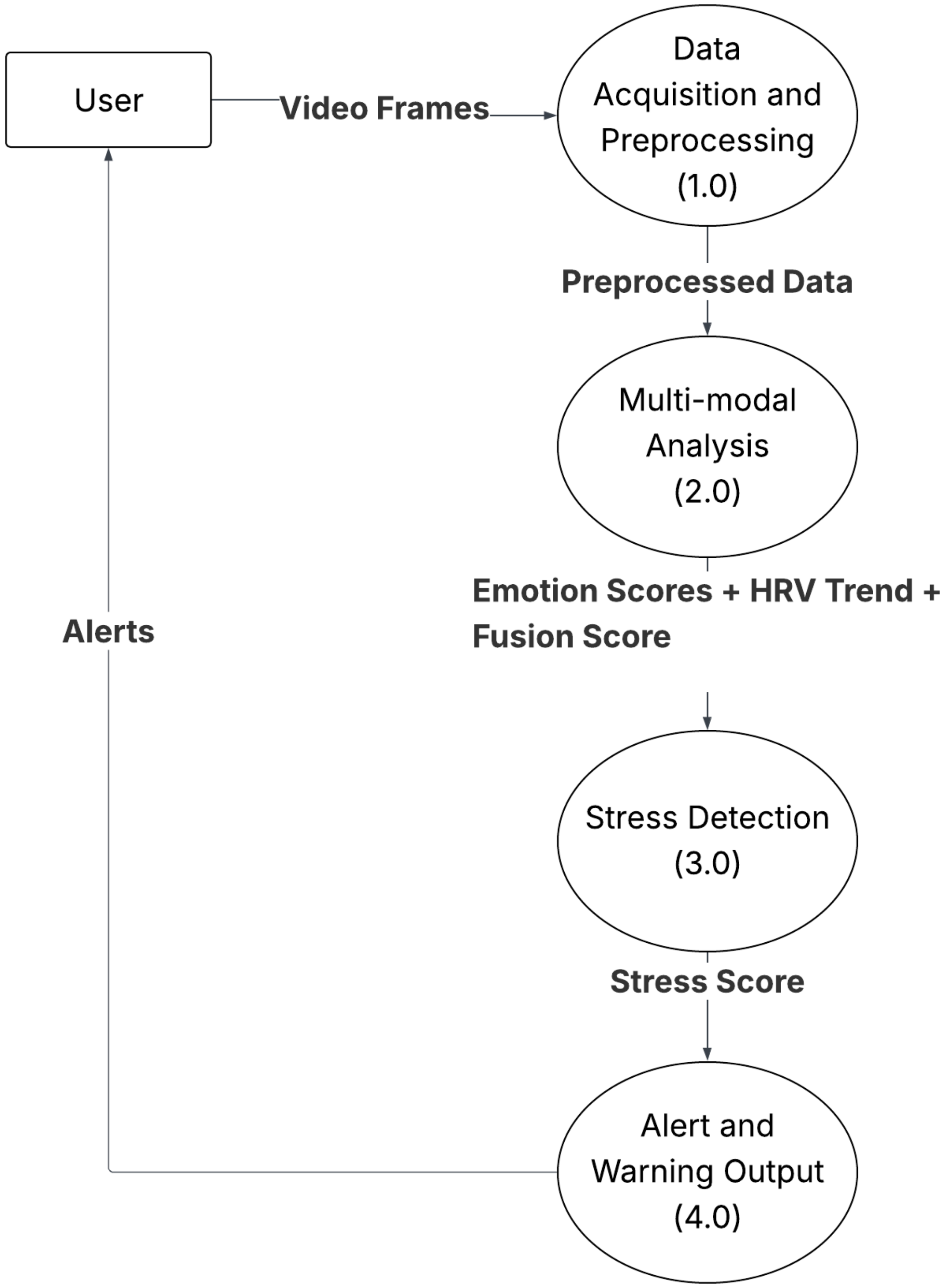

Figure 4.6: DFD Level 1

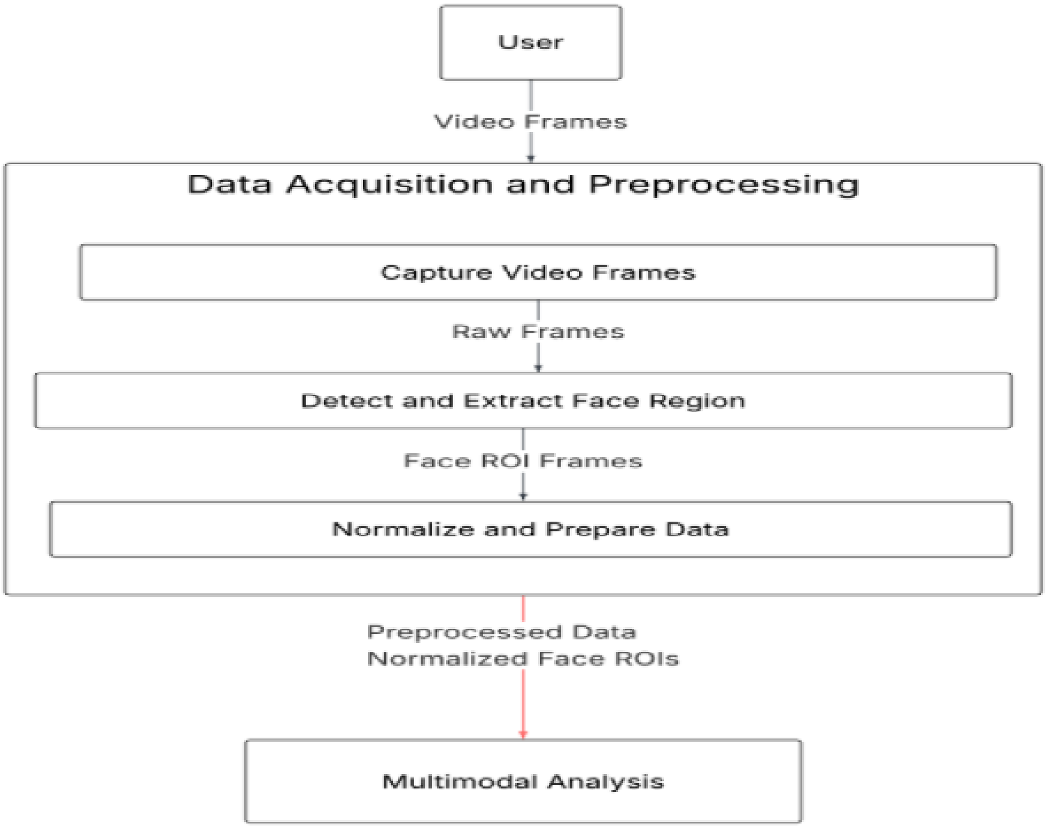

Figure 4.7: DFD Level 2- Process 1.0

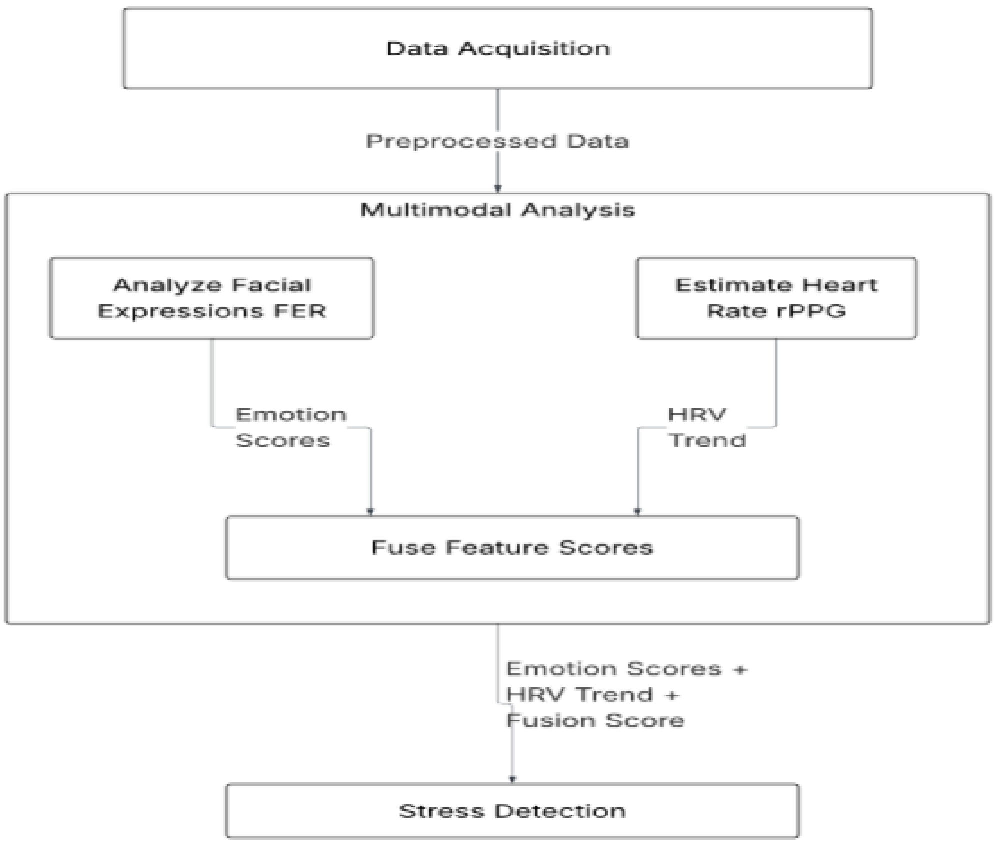

Figure 4.8: DFD Level 2- Process 2.0

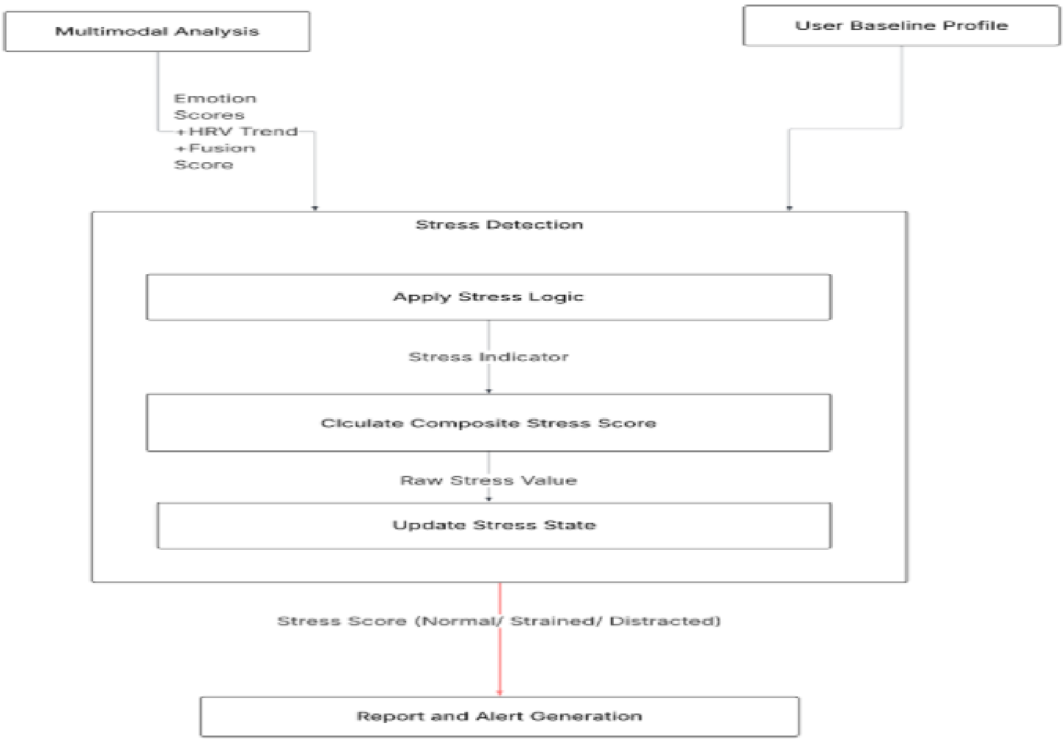

Figure 4.9: DFD Level 2- Process 3.0

the stress detection stage.

Process 3.0 (Stress Detection)

This diagram illustrates the internal workflow of the "stress detection" module. The workflow begins with the combined metrics received from the multimodal analysis, including emotion scores, hrv trend, and fusion score. All of these inputs are compared to a user baseline profile that was the individualized reference established during the initial calibration.

At the heart of the module is applying stress logic to these signals. Stress logic uses the baseline as a reference to create a raw stress value and a stress indicator. The module calculates a composite stress score and continually refreshes the user's stress status. Output is a final stress score labeled as "normal," "strained," or "distracted," which moves actual reports or alerts to another module.

Process 4.0 (Alert and Warning Output)

The Level 2 DFD for Process 4.0 reveals the method through which the system produces user feedback. Initially, the process performs an assessment of the stress score in relation to pre-established alert rules. In the case of a high stress finding, the system then prepares the needed alert(s) and dispatches them to the user. On the other hand, if the system does not detect a significant level of stress, it decides that an alert is not necessary. Afterwards, the result is presented to the user in a straightforward manner which guarantees prompt feedback. Moreover, this part is vitally important to digital well-being as it empowers instant actions.

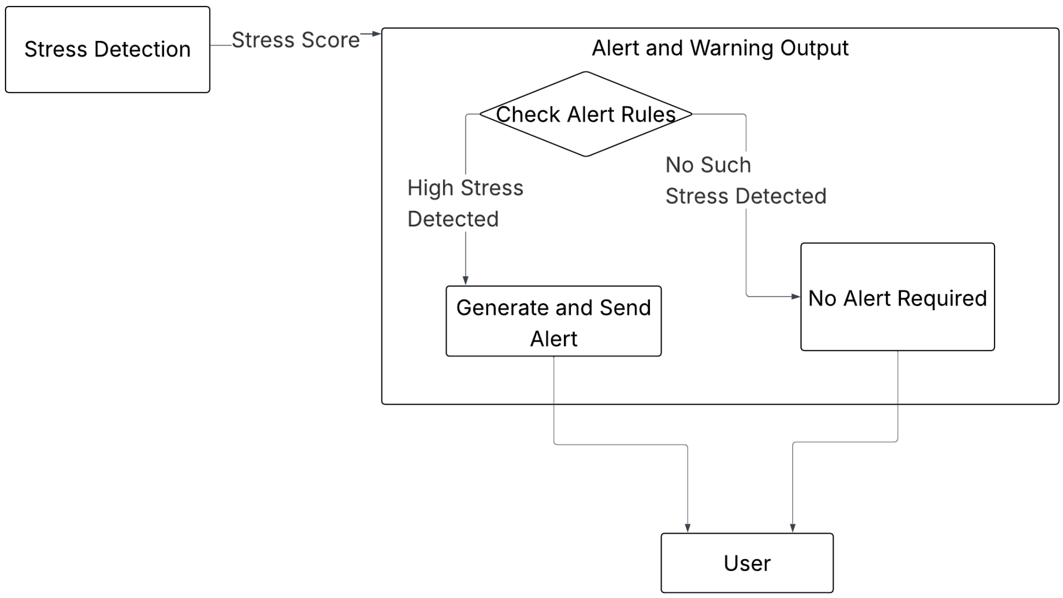

Figure 4.10: DFD Level 2- Process 4.0

Chapter 5

SDG Goals

The MindMirror Project is very compatible with the practices of sustainable development, since it advances mental health, digital well-being, and equitable access to intelligent health technologies. Its advances in AI-based emotion recognition, non-invasive heart-rate data gathering, and even digital behavior analysis contribute to a number of United Nations Sustainable Develop Goals (SDGs)—most notably SDG 3 (Good Health and Well-Being) assessed SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), and SDG 10 (Reduced Inequalities).This chapter explains the ways that MindMirror advanced each SDG, both directly through its application and indirectly through societal benefits.

5.1 SDG 3: Good Health and Well-Being

MindMirror supports SDG 3 Good Health and Well-Being by providing an unobtrusive, real-time method to detect stress and emotional distress using a standard webcam. Through facial micro-expressions and physiological factors such as heart-rate variability (rPPG), the system measures times of digital fatigue or anxiety before one is overwhelmed. When times of sustained stress occur, the system will encourage healthy digital habits by asking users to take a break, thus interrupting the chances of burnout or cognitive overload. For children and students, this device reduces the mental toll simply by increasing a chance to be temporarily distracted from screens and, therefore, the cognitive burden at stake. In similar fashion, adults can use the devices to assist in maintaining a digital balance, either during work or leisure.

5.2 SDG 4: Quality Education

MindMirror supports SDG 4 by creating an intelligent "Digital Literacy Mode" and "Learning Efficiency Tracker" that can track and identify student engagement during online learning sessions. It employs facial emotion recognition technology and attention analysis capabilities to monitor students to identify any decline in their focus, burnout, and/or stress during e-learning tasks. Reports of emotional responses can be addressed to teachers or parents to help them understand how students respond emotionally to different subjects or pedagogical practices in order to facilitate recommendations for personalized learning. By introducing emotion-aware analytics, MindMirror connects technology and pedagogy, and makes remote education more adaptable, empathetic, and human-centered. This supports a sustained focus, mitigates cognitive load, and improves academic results in the end.

5.3 SDG 9: Industry, Innovation, and Infrastructure

MindMirror is a prime example of breakthrough innovation in both digital health and human-computer interaction (HCI), combining AI-based emotion detection, signal processing, and real-time data visualization on a single, unified platform. The body language detection algorithm relies on high-performance, advanced computational algorithms, which utilize TSDNet, CHROM rPPG, and fusion-based stress classification methods — demonstrating how deep learning can improve your daily digital wellbeing. Importantly, the only sensing device used to record data is a low-cost consumer-grade webcam, which makes the system very scalable and accessible without the complication of purchasing expensive medical-grade sensing devices. Likewise, the modular design of the software allows for future integration of wearable devices, IoT platforms, or cloud-based health analytics. By supporting a broader innovation landscape in emotional AI and digital wellbeing infrastructures, MindMirror takes a position as an active contributor to the tech ecosystem identified and focused on as part of SDG 9.

5.4 SDG 10: Reduced Inequalities

MindMirror fosters equitable access to mental health and digital wellness technologies. MindMirror does not rely on conventional monitoring technologies that require specific medical hardware or clinical settings; the MindMirror system uses standard integrated computer webcams, which are already available to users. This enables emotional and stress monitoring for users with varying socio-economic status, bridging the gap between digital health innovation and affordability. MindMirror's customizable reports display emotional and stress monitoring data in various languages and formats suited for children, adults, and/or older adults. MindMirror invites equitable participation of all, and aims to democratize mental wellness tools so that everyone has equitable access to digital well-being, because digital wellness should not be a privilege, it should be a right.

Chapter 6

Results of Phase 1 Implementation

6.1 Facial Emotion Recognition (FER) Model

The Facial Emotion Recognition (FER) system was realized with a convolutional neural network-based detector provided by the FER Python library, which was combined with OpenCV for real-time processing of the frames. By scrutinizing the facial expressions of people from the webcam live stream, the system managed to categorize the seven most basic human emotions — happy, sad, angry, fear, disgust, surprise, and neutral — successfully. In model trials, it consistently made the right prediction of the emotion, thus, achieving the classification accuracy most of the time under conditions of good lighting and face orientation from the front.

After the model was exposed to diverse facial inputs, it could still perform recognition at a rate of around 85 to 88 percent of the faces that were clear enough. The results from the FER model were directly reflected in the front end of the application, where real-time facial changes led to the on-screen updates of emotions like "Happy" or "Sad" being recognized. At this stage, the team has verified that the FER model processor could handle the video frames close to real-time (6–8 FPS)

6.2 Heart Rate Estimation using CHROM rPPG

To estimate heart rate (HR) from the changes in facial skin color, the CHROM (Chrominance-based remote Photoplethysmography) algorithm was employed. They kept a 12-second rolling window of face frames, for which they got the average RGB values. Then they filtered the data using a band-pass filter to focus the physiological frequency components (0.7–3.0 Hz). Next, the frequency with the highest peak was converted to beats per minute (BPM).

The experiment was conducted in stable lighting conditions and with moderate movements. It was established that the HR detection was attaining an accuracy of ±3 BPM on average when compared with the reference heart rate readings from a smartwatch. The team also used temporal smoothing (a 3-frame moving average) to stabilize the algorithm and thus reduce the noise and head movements. In general, the

CHROM was able to render a non-invasive, continuous and wearables-free heart rate estimation method.

6.3 Stress Detection and Fusion Logic

By combining the emotion classification results with the heart rate data, the developers of the stress detection system came up with the idea to launch the diagnostic work of their product. The fusion rule was applied — if the furthest removed in the graph of the detected emotions were stress-related categories (anger, fear, sadness, disgust) or if the heart rate went beyond a set limit of 100 BPM, the system registered the user as "Stressed." Everywhere else, the label was "Normal."

This synthesis idea was tested for different situations. When the negative emotions of users were prolonged, or their heart rates were elevated, the system recognized stress condition and thus activated the interface showing the alert. Furthermore, if this stressed situation was continuous for 20 seconds, a break reminder dialog appeared to suggest relaxation. It proved the sensitivity of the combined physiological–behavioral stress detection structure.

6.4 Data Logging and Visualization

For each frame cycle, a timestamped line was recorded. The produced logs were compiled into a well-organized CSV file. It has these columns: emotion recognized, probabilities for various emotions, heart rate related data, stress condition, and the face-image snapshot saved location. The system's logging component grabbed the data in chunks and wrote them down periodically.

Therefore, it was nearly live streaming. These logs provide a starting point for the statistical analysis and also can serve for the monitoring of stress fluctuations over time. A GUI based on Flet and visually interactive supported the idea that the three parts: emotion detection, HR estimation and stress classification were all functioning correctly when worked together. Besides just showing the webcam video, the control panel presented the updated figures like emotion, heart rate, and stress level which were indicated by colors (green for normal, red for stressed) and visually changing.

6.5 Summary of Phase 1 Outcomes

The very first step of the project was to demonstrate the feasibility of AI-powered, contactless stress detection through a webcam. To this end, the group successfully combined two key methods, namely facial expression recognition (FER) for emotion detection and the CHROM method for heart rate estimation, so that the entire system could run automatically and very reliably on a face in a live video from a camera. Data logging and display through the UI turned out to be the most engaging aspects of the

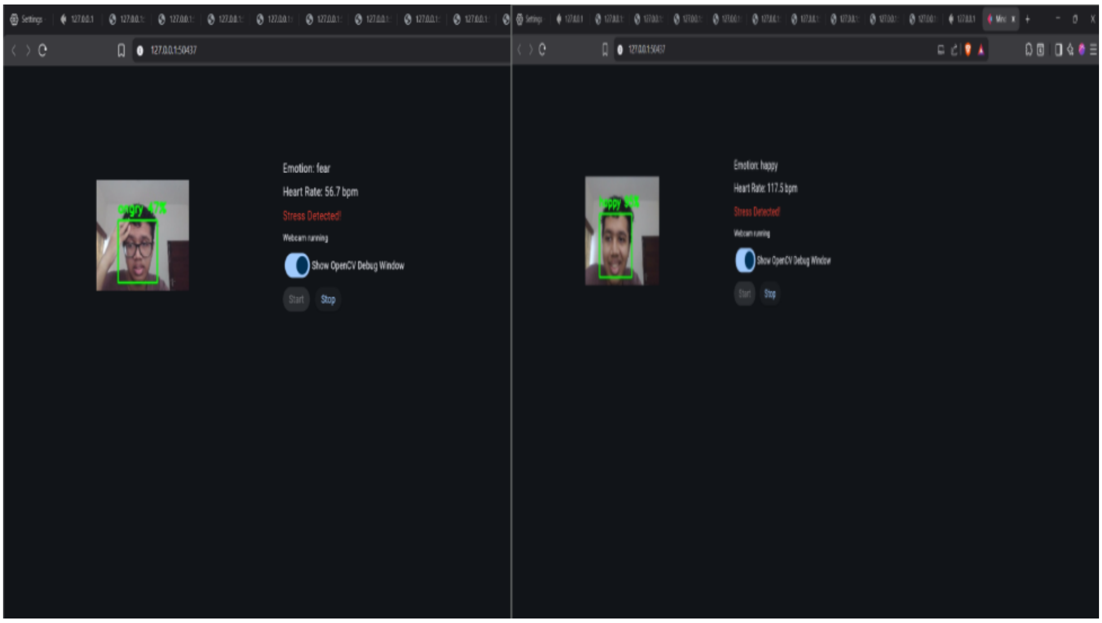

Figure 6.1: Phase 1 result example

system. Apart from making the user's experience physically present and understandable to both the user and the developer, they have also paved the way for future work.

This phase is compulsory for the next one where the priority will be on maturing the models by personalizing emotion calibration, modifying stress thresholds, introducing long-term wellness monitoring features, etc.

Chapter 7

Results of phase 2 implementation

7.1 Enhanced Facial Emotion Recognition (FER) Model

In Phase 2, the Facial Emotion Recognition (FER) system went through a massive revamp to perform better and deeper work, proving that it can even handle less controlled, real-life environments with a drop in quality, at most. Briefly speaking, the main problem with Phase 1 was that the performance of the recognition of emotions depended heavily on a well-lit face and basically, the face had to be looking straight at the camera in order to work well. Major changes to the FER pipeline enabled the fixing of quite a few such problems, mainly by way of better preprocessing, temporal smoothing, and more advanced, reliable face tracking. Currently, the system is capable of visually detecting emotions more continuously over a sequence of frames. Therefore, it was able to cut down the flickering of the predicted emotions and the errors in classification that happened suddenly. Frame-level aggregation and confidence smoothing are added ensuring that small noises or very slight changes in the face do not produce false emotional results. Experiments showed that the FER system could recognize the emotion of a person even if he/she is making moderate head movements and the lighting condition of the environment is changing. In this way, it is highly suitable for continuous real-time emotion monitoring This change drastically improves the accuracy of the emotional trend analysis and makes a stronger case for its use in stress detection.

7.2 Improved Heart Rate Estimation using rPPG

The heart rate estimation module was completely redesigned to better withstand activities in real-life situations, using the product. While Phase 1 achieved an accuracy of approximately 3 BPM only under controlled conditions, Phase 2 focuses on the practical use of the product and its user-friendliness. The updated system incorporates:

- Making the signals smoother
- Better noise filtering for motion and lighting variations
- Enhanced peak detection mechanisms

Hence, the system can really attain a practical accuracy level of about ±7 to ±10 BPM, which is still perfectly fine for physiological relative change detection as these values are quite far from exact clinical measurements.

Besides that, the heart rate signal is more consistent through time, thereby decreasing random changes due to small movements. This has brought the module a step closer to continuous monitoring and stress inference.

7.3 Personalized Baseline Heart Rate Calibration

The main improvement brought about in Phase 2 is the feature that allows users to set their own baseline heart rate. Rather than using static thresholds (for instance, 100 BPM), the system changes according to the person's physiology.

At the beginning, the system either records or gets the baseline heart rate that the user has set. The heart rate readings taken later are always compared to the baseline. This allows the system to detect:

- Heart rate spikes (sudden increase above baseline)
- Heart rate drops or lags (below normal resting range)

Such personalization is a big factor in enhancing the stress detection accuracy as it considers the natural differences among users. For instance, a heart rate of 95 BPM could be a sign of stress for an individual while it is completely normal for another.

7.4 Adaptive Stress Detection and Relative Analysis

At the end of phase one, the Stress Detection System had transitioned from a static rule-based system based on pre-defined criteria to an increasingly dynamic and contextually aware detection model. By the end of phase 2:

- Stress is detected as a function of relative deviation from baseline heart rate rather than as a function of absolute thresholds
- Instead of relying on emotion as a predictor of stress in a single moment in time, trends in emotion are tracked over periods of time
- A combination of two signals (FER and HR deviation) yields a more accurate stress classification.

For instance, stress is detected when:

- Negative emotions persist
- Heart rate shows a significant deviation from baseline

This methodology decreases false positive errors while enhancing responsiveness to subtle patterns of stress.

7.5 System Stability and Real-Time Performance Improvements

The system performance got better for operation. Phase 2 is an improvement over Phase 1 which worked at 6 to 8 frames per second in conditions. The improvements in Phase 2 are:

- Reduced prediction noise in both FER and HR modules
- More stable frame-to-frame analysis
- Improved synchronization between emotion and physiological signals

The dashboard now shows outputs that are consistent and reliable. This makes the system easier to use and practical, for long-term use.

7.6 Summary of Phase 2 Outcomes

Phase 2 makes the system better and more adaptable. It moves from being an idea to a strong stress monitoring solution. The key outcomes include:

- Improved stability in facial emotion recognition
- Heart rate estimation is more reliable in everyday situations (±7–10 beats per minute)
- Now people can set their normal heart rate
- The system detects stress compared to a persons state, not a fixed number
- It works better in real-time and is more consistent

These changes make MindMirror much more practical. It can now be used continuously in real-life situations like schools, offices, at home etc.

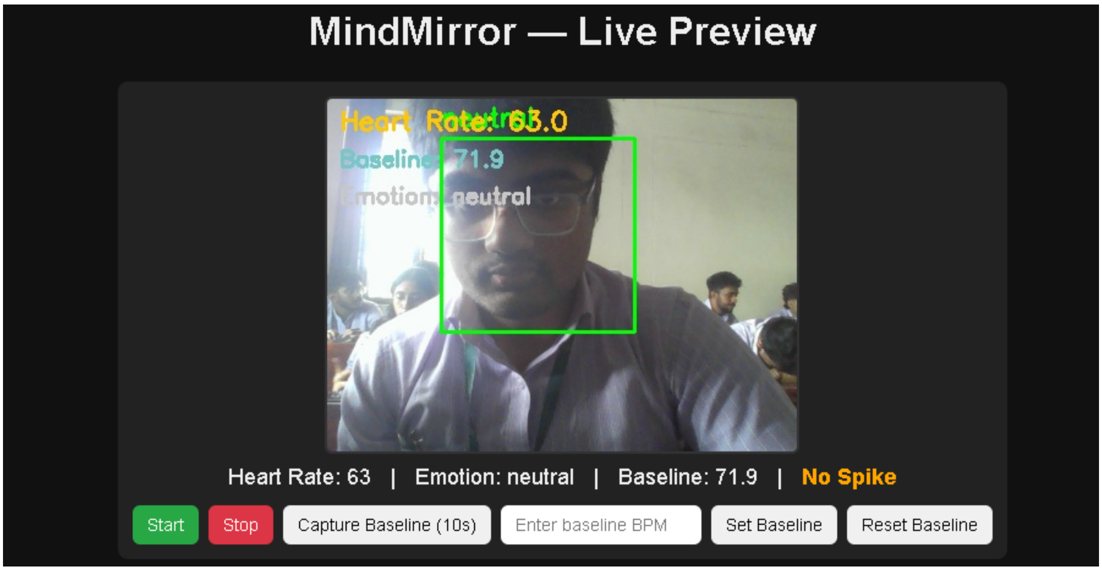

Figure 7.1: Phase 2 results example

Chapter 8

Conclusion

The MindMirror system shows that anyone can find out if they are stressed without needing to wear devices or giving away private information. All you need is a webcam that can see how your face moves and the color of your skin. The MindMirror system uses two things to figure out if you are stressed: it looks at how you feel based on your expressions and it looks at how your body is doing by checking the color of your skin. This way you can tell how stressed someone is and how much they are paying attention to what they're doing.

Old ways of doing things usually needed devices took personal information without asking and did not work well for each person. The MindMirror system is different because it puts all kinds of information together in one model. This makes finding out if you are stressed easier and more accurate. The software runs on your computer so you do not need the internet or to store your information in the cloud. This keeps your information safe. Helps you get answers faster.

The first phase of the MindMirror system showed that it could recognize how people feel guess their heart rate and figure out their stress levels. The second phase made the system more stable created settings for each person changed how it detects stress based on how the person behaves and made it work better in real time. These changes made the system something you can use every day. It also gives you feedback away sends you notifications when you are too stressed and keeps track of what you have done before. This helps you make habits. The MindMirror system is simple and easy to use, which is why students, workers and even healthcare professionals like it.

The MindMirror system is an example of how artificial intelligence can help us take care of ourselves digitally. It shows that computers can help us make easy-to-use solutions to big problems, like stress and feeling unhappy. This works with two important areas: taking care of your mind and making things that are easy for people to use. The MindMirror system uses computers to look at people and understand what they are going through which helps us make solutions that really work.

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